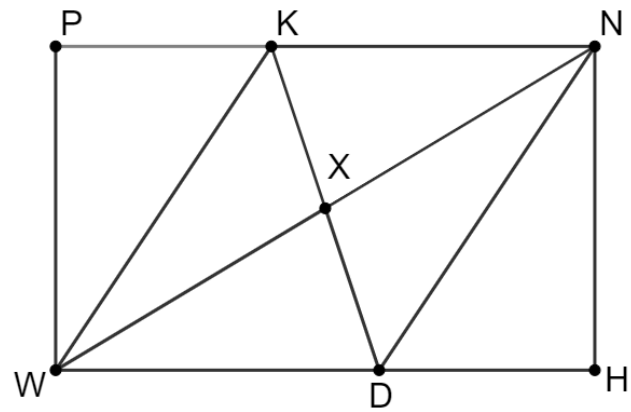
Geometry Name____________________
Unit 5 Date____________________
Lesson 8 Practice Period__________________

Quadrilateral $PNHW$ is shown where $\overline{KW} \parallel \overline{ND}$ and $X$ is the midpoint of $\overline{WN}$

Given: $\overline{KW} \parallel \overline{ND}$, $X$ is the midpoint of $\overline{WN}$
Prove: $\overline{WK} \cong \overline{DN}$




Use the given information to answer the questions about the incomplete proof that is shown.

| **Statement** | **Reason** |
|---|---|
| 1. $\overline{KW} \parallel \overline{ND}$ | 1. Given |
| 2. | 2. |
| 3. $X$ is the midpoint of $\overline{WN}$ | 3. Given |
| 4. | 4. |
| 5. | 5. |
| 6. | 6. |
| 7. $\overline{WK} \cong \overline{DN}$ | 7. |

1. What is the statement for line 2 of the proof?

2. What is the reason for line 2 of the proof?

3. What is the statement for line 4 of the proof?

4. What is the reason for line 4 of the proof?

5. What is the statement for line 5 of the proof?

6. What is the reason for line 5 of the proof?

7. What is the statement for line 6 of the proof?

8. What is the reason for line 6 of the proof?

9. What is the reason for line 7 of the proof?

10. Explain if it is possible to prove that $\triangle WXD \cong \triangle NXK$ using the same given statements.